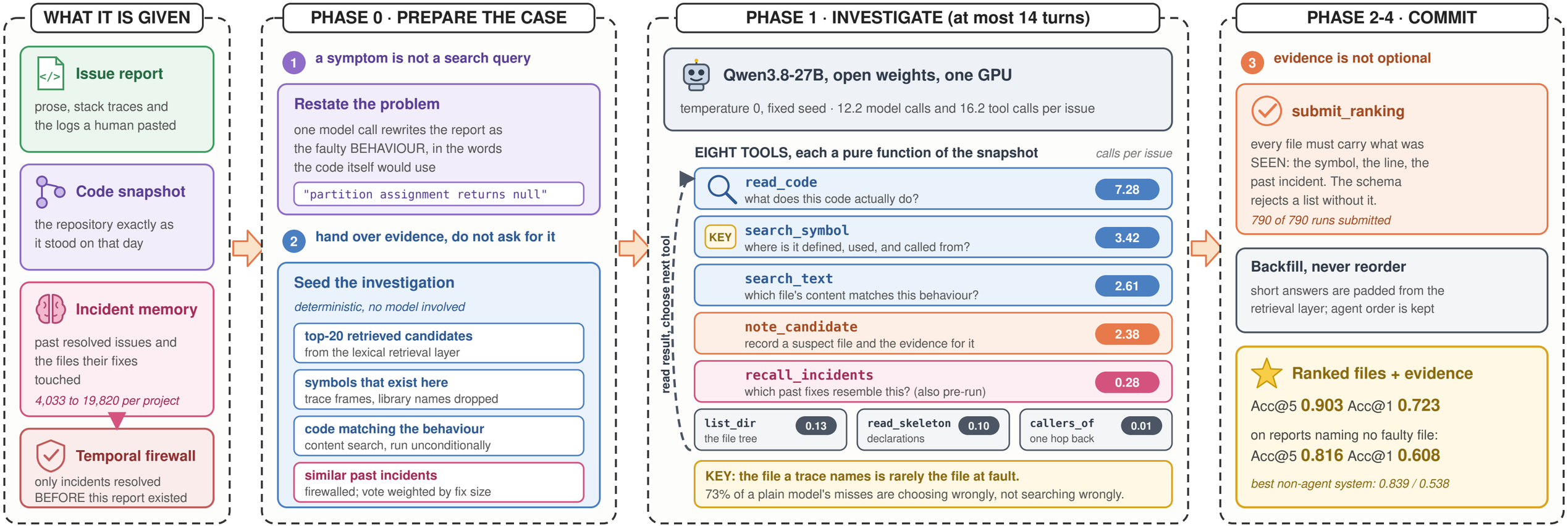

WHAT IT IS GIVEN
Issue report
prose, stack traces and the logs a human pasted
Code snapshot
the repository exactly as it stood on that day
Incident memory
past resolved issues and the files their fixes touched
4,033 to 19,820 per project
Temporal firewall
only incidents resolved BEFORE this report existed
PHASE 0 · PREPARE THE CASE
1 a symptom is not a search query
Restate the problem
one model call rewrites the report as the faulty BEHAVIOUR, in the words the code itself would use
"partition assignment returns null"
2 hand over evidence, do not ask for it
Seed the investigation
deterministic, no model involved
top-20 retrieved candidates
from the lexical retrieval layer
symbols that exist here
trace frames, library names dropped
code matching the behaviour
content search, run unconditionally
similar past incidents
firewalled; vote weighted by fix size
PHASE 1 · INVESTIGATE (at most 14 turns)
Qwen3.8-27B, open weights, one GPU
temperature 0, fixed seed · 12.2 model calls and 16.2 tool calls per issue
EIGHT TOOLS, each a pure function of the snapshot
calls per issue
read result, choose next tool
read_code
what does this code actually do?
7.28
KEY
search_symbol
where is it defined, used, and called from?
3.42
search_text
which file's content matches this behaviour?
2.61
note_candidate
record a suspect file and the evidence for it
2.38
recall_incidents
which past fixes resemble this? (also pre-run)
0.28
list_dir
the file tree
0.13
read_skeleton
declarations
0.10
callers_of
one hop back
0.01
KEY: the file a trace names is rarely the file at fault.
73% of a plain model's misses are choosing wrongly, not searching wrongly.
PHASE 2-4 · COMMIT
3 evidence is not optional
submit_ranking
every file must carry what was SEEN: the symbol, the line, the past incident. The schema rejects a list without it.
790 of 790 runs submitted
Backfill, never reorder
short answers are padded from the retrieval layer; agent order is kept
Ranked files + evidence
Acc@5 0.903 Acc@1 0.723
on reports naming no faulty file:
Acc@5 0.816 Acc@1 0.608
best non-agent system: 0.839 / 0.538